Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer — Apr 2023 - Present

Interconnect Architect | CMN GUI Lead | Tools for internal engineers and external customers.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025), for a project-agnostic multi-agent orchestration and context-gathering system.

- Created all coherent-mesh (CMN) GUI tooling from CMN-800 onward, including Multi-CHI port devices, MCS group configuration, custom routing and mesh deadlock detection interfaces.
- Designed a modular AI engineering platform with four project packs, separating shared tooling from product-owned runtimes through manifest-driven configuration.
- Built a Python NIC-400 XML/CCXML migration engine that reconstructs topology and routes while preserving placement; historical validation recorded 4,130/4,130 valid configurations passing design-rule checks and 5/5 curated invalid configurations rejected.
- Designed and implemented Python tooling using Model Context Protocol (MCP) for two bounded interconnect construction workflows, with typed intent, dependency-ordered plans, checkpointed execution, semantic validation and rollback.
- Built a Rust/Tauri 2 workbench with React/TypeScript UI, a shared desktop/CLI core, bounded asynchronous Git inspection and verified process controls.
- Created the NCI/NoC logical canvas and console for configuration logs and design-rule results; built canvas exports and configuration-report scripts.

Software Engineer — Apr 2020 - Apr 2023

- Implemented Flask services to start and stop Jupyter notebooks and persist files in AWS EFS for Pelion's embedded-ML platform; defined PostgreSQL schemas and built end-to-end tests.
- Built sensor-ingestion and dashboard APIs with InfluxDB, including a fixed-radius neighbor algorithm for workspace proximity detection.

Graduate Software Engineer — Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware for I2C, GPIO, ADC, DAC and PWM; contributed to shield/form-factor PCBs. 2020 Arm Quality Award (team).
- Built Node.js/Express REST APIs, PostgreSQL schemas and React dashboards to monitor occupancy from thermal cameras and PIR sensors.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust, SQL, Verilog
Applications: Angular, Electron, React, Tauri 2, Flask, Node.js, SVG/Pixi
Engineering: MCP, Jenkins, Playwright, Git, LSF; SoC/interconnects, AMBA, FPGA, Mbed OS

EDUCATION

The University of Texas at Austin

Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits
1st Place (Industry), ECE Capstone Design Award - IoT Wireless Home Security System (2018)